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DOI: 10.1109/FG47880.2020.00086

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A weightless regression system for predicting multi-modal empathy

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***Abstract*— This work takes into account the benefits of machine learning in order to estimate the valence of emotions on the OMG Empathy dataset, considering the information obtained from face expressions and dialogue of interlocutors. RegressionWiSARD and ClusRegressionWiSARD n-tuple regressors and its ensembles were employed to this end. The best performance achieved among all the combinations of weightless neural models considered (evaluated using the CCC metric) was 0.25 in validation set of the Personalized Track .**

I. Introduction

Since emotional states are a fundamental part of the core of human psychology, often exceeding the intellect itself in psychological hierarchy, it is natural that Affective Computing[11] occupies a prominent place in the study of the human-machine interface. Along with the apogee of machine learning, Affective Computing has experienced great growth in recent years, but it still has many open questions. Some of the main ones involve the prediction of emotions based on information from many different sources and the identification of subtle emotional states in real time. Specifically, many advances have been made recently in the area of affective prediction[25][26][27][28][29][30][31][32].

In order to offer a significant contribution in the area, this paper discusses the use of weightless neural network ensembles in predicting the affective valence of individuals in conversation videos, since this type of model has computational simplicity, great computational agility and ease of being parallelized.

The structure of this work is as follows: in Section 2 the WiSARD weightless model, some of its extensions and ensembles will be described, Section 3 deals with the preprocessing of data from different sources, Section 4 describes the experiments carried out with different types of ensembles using uni and multimodal data and discusses their results, and Section 5 is the conclusion of this work, summarizing all the previous discussion and also offering the main ongoing works.

II. n-Tuple models

The n-Tuple classifier is a boolean node pattern classifier [3], which distances itself from models derived from perceptron because it do not use synaptic weights between their neurons, thus avoiding all training time required for their convergence. n-Tuple classifier does not need any parameter fine tuning, nor does it use any error minimization technique to obtain generalization in pattern learning [17]. The family of models derived from the n-Tuple classifier is known as Weightless Artificial Neural Network (WANN).

This work was not supported by any organization

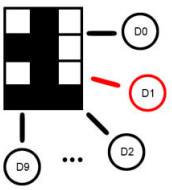


Fig. 1. The WiSARD model multidiscriminator structure. For digits recognition task there are ten discriminators. In the training phase, only the corresponding discriminator is accessed.

A. WiSARD

WiSARD is a neural model based on the n-tuple classifier, where each neuron is equivalent to a piece of memory [1]. This model is class discriminator-oriented, where all discriminators are formed by N RAM-neurons, whose memory addresses are addressed by n-bit tuples. Each neuron has 2^n memory locations.

WiSARD works with binary standards, requiring the use of some preprocessing technique to form data suitable to the model before the training and classification process. The training process consists of using the binary input to access specific memory positions of the corresponding discriminator and increment the counter that constitutes its content. During the classification, all discriminators are accessed and they are assigned a score formed by the number of non-null positions accessed. The discriminator with the highest score will determine the class of the entry and in case of a tie, a threshold called bleaching, which is initialized to zero, is increased and the classification is repeated, considering for the score only memory locations whose counter has higher value than bleaching. This procedure is repeated until there is a winning discriminator or until the bleaching value exceeds the highest counter among the memory locations accessed, in which case a default class is chosen for the entry. The structure and the training process in WiSARD are illustrated in Figs. 1 and 2. WiSARD can be used to accelerate the training of deep models, and can be used as a starting layer for such neural networks in a hybrid hierarchy[33].

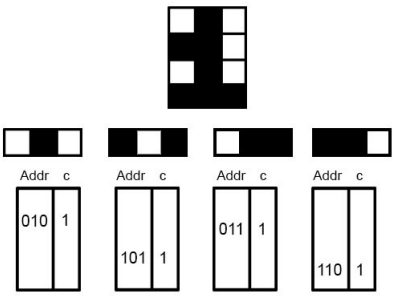


Fig. 2. Training a WiSARD discriminator. The input is divided into N tuples formed by a random selection of input bits.

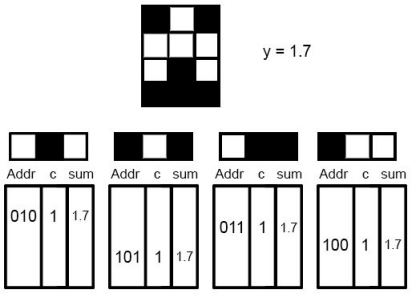


Fig. 3. RegressionWiSARD extends the WiSARD model by adding and extra dimension to the RAM nodes, which represents the sum of all training observations correct outputs.

B. ClusWiSARD

ClusWiSARD is an extension of the original WiSARD model for handling datasets that include very different examples in their classes, separating them into subprofiles or clusters [7]. ClusWiSARD was created based on the Plasticity-stability dilemma, which seeks to understand if a particular cluster is still able to receive information and expand its diameter, without harming the homogeneity of your data. The intuition behind this new model lies in the ability of a discriminator to recognize that an example of his class is not sufficiently similar to those already learned and the network to create in dinamically way a new discriminator for this same class that can absorb this rejected example. In this way, WiSARD could simultaneously handle supervised, semi-supervised and unsupervised learning, suffice it for this that an unlabeled date is trained on the discriminator with the highest score.

C. Regression WiSARD

Regression WiSARD (ReW) is an extension to WiSARD, based on the n-tuple Regression Network, a previous regression WANN [4]. In this model, the content of the RAMs is an access counter and a prediction field. When the model is trained with a pair $< \mathbf{x}, y >$, the binary input $\mathbf{x}$ is used to access a specific memory position and increment its counter and the associated prediction field is increased by the value of y, which is a real value, as exemplified in Fig. 3. Since there is no classification, ReW does not use discriminators. During its prediction phase, a binary input is submitted to the network and is used to access the corresponding memory locations. The value of counters and accessed partial predictions are used to calculate an average (simple mean, power mean, median, harmonic mean, harmonic power mean, geometric mean and exponential mean), which will be the final prediction value.

A version of ReW inspired by ClusWiSARD was also created and is known as ClusRegression WiSARD (CReW). This model is initialized with only one ReW discriminator, but if any example presented for training is not sufficiently similar to those already learned by this ReW, a new discriminator is created. During the training phase, each example can be learned by more than one ReW and the discriminators who will learn the example are chosen through a simple classification. During the prediction phase, a classification is also performed to select which ReW will perform the operation.

D. Ensemble learning

In order to obtain a more accurate prediction, ReWs and CReWs are grouped using ensemble learning, where a set of models, here called weak learners, is generated and, generally, each of them gets specialized training. The final prediction is an average of the weak learners' predictions. In the case of Regression WiSARD, three types of ensembles are traditionally used:

- Bagging: each weak learner is trained using a part of the training set, with replacement [15];
- Boost: each weak learner is trained using a part of the training set, without replacement. A validation phase is performed after the end of the training, where the prediction of each weak learner receives a weight associated with its performance in this validation. These weights are used to calculate the influence of each weak learner on the final prediction value [14];
- Naïve: each weak learner is trained with the entire dataset.

In these ensembles, the parameters of each weak learner are drawn, and each network can use a different average when calculating partial predictions. The ensembles can be homogeneous, using only ReW or only CReW, or heterogeneous, mixing the models.

E. KernelCanvas

KernelCanvas[19] is a specific type of preprocessing to turn WiSARD able to deal with time-series classification. Since WiSARD only works with data of equal sizes, a craft is required when they are in different sizes, which usually occurs with time series. The inspiration for this idea comes on the action of using a paintbrush on a white canvas. When drawing a figure on the canvas, it does not matter how many times the brush stroke the canvas or the order the figure was

drawn, since the canvas itself will preserve its dimensions. In a similar way, KernelCanvas makes use of a set of kernels to represent partitions of the canvas.from the fact that you can colorize a region that includes a point, thus discretizing the input space and reducing its dimensionality by having a bit group replaced by the bit representing its region.

III. Preprocessing techniques

Since WiSARD-based models can handle only binary inputs, some type of preprocessing is necessary to make input suitable for the network. Here the binarizations used in this work will be described.

A. Images

Adaptive Gaussian[22] is a low-pass filter that binarizes the image according to a local threshold for each pixel. Sauvola method [20] binarizes a integral image, i.e., a version of the image in which every pixel has its value substituted by the sum of the values of every pixel that lie to left and/or above it. For performs this binarization, Sauvola employs this threshold:

$$t = 1 + \mu_F + k\left(\frac{\sigma_F}{r} - 1\right), \tag{1}$$

where F represents the features of the neighboring pixels, μ_F and σ_F their respective mean and standard deviation, and k and r are factors .

The Canny border detector[21] consists of smoothing the image, applying a Gaussian filter, calculating its intensity gradients and then removing spurious edges using non-maximum suppression techniques. After that, two thresholds are applied to the image. A pixel whose intensity gradient is higher than is maintained in binarization, one whose intensity is lower and removed, and those between the two thresholds are only maintained only if they are connected to a strong edge.

Otsu's binarization[23] employs a clustering approach, separating pixels from grayscale images using a bimodal histogram. This binarization achieves an optimal threshold that separates these pixels through the minimization of intraclass variance.

B. Audio

The only kind of audio preprocessing used here was mel-frequency cepstral coefficient extraction[24]. Mel-frequency cepstrum (MFC) is a representation of the short-term power spectrum of a sound and MFCC are coefficients that collectively make up an MFC. Essentially here an MFCC is obtained by applying a Fourier transform to the audio signal, and then using the integral image of MFCC within a pre-specified context window. Then the input is normalized according to the training set MFCCs. Since audio signals have different sizes, the input is processed by KernelCanvas.

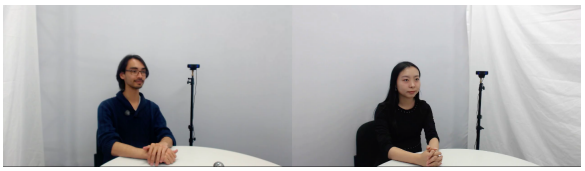


Fig. 4. Example of OMG-Empathy dataset video.

IV. Experimental results

A. Experimental setup

The dataset used to validate WiSARD's capability to predict emotional value in dialogues was the OMG-Empathy Dataset[18]. The dataset is composed of 80 videos, each lasting approximately 5 minutes, in which actors tell stories to listeners, who then evaluated their valence level with a continuous value between -1 and 1 at each point in the conversation. The challenge proposed for this dataset has two tracks: Personalized, where only the valency of the listerner's emotion is taken into account, and Generalized, where the valence to be predicted uses information from both listener and actor. A example of OMG video is displayed in Figure 4.

The metric used here for measures the quality of the empathy's prediction is concordance correlation coefficient (CCC) ρ_c. CCC is given by

$$\rho_c = \frac{2\rho\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2 + (\mu_x - \mu_y)^2}, \tag{2}$$

where μ_x and μ_y are the means of x and y, σ_x and σ_y their standard deviations, and ρ is the Pearson's correlation coefficient between both variables.

Two types of experiments were performed here, both on Personalized Track: multi-modal prediction using a single network, ReW and CReW, in validation set and uni-modal using heterogeneous Bagging and Naïve ensembles of Regression WiSARD, with only five weak learners each, using simple mean of the individual learners results to generate the ensemble result in validation set. The weak learners are ReWs and CReWs, whose address sizes vary randomly between 5 and 32 bits and the average of each network is drawn among the whole set of possibilities that the model offers. The multi-modal input is formed by the simple concatenation of the binarization of the image of the listerner's face in the frame whose valence is desired to be predicted with the MFCC obtained from an audio window centered on this same frame. This window extends from 30s before the frame to 30s after it, and such information cannot even exist, so that even the size of the audio data is variable. And the uni-modal consist only in facial image.

The experimental environment was an Intel Core i7-6700, CPU @ 3.40GHz with 4 cores, 32GB of RAM and GeForce GTX 960. The WANN implementations are those available in the C++/Python library wisardpkg.

B. Analysis of results

All the results now presented are from experiments still in progress. Table I shows the average time of each prepro-

cessing method.

Preprocessing	Average time (s)
Sauvola	0.0004
Canny	0.0002
Gaussian	0.001
Otsu	0.00005
MFCC	10.54

TABLE I

AVERAGE TIME OF EACH PREPROCESSING METHOD.

Tables II and III show the results in the validation set for ReW and CReW, respectively.

	n					
Subject	5	10	15	20	25	30
2	0.09	0.08	0.04	-0.28	**0.6**	0.34
3	-0.02	-0.01	0.04	0.17	0.36	**0.38**
4	-0.25	-0.19	-0.07	0.19	0.44	**0.45**
6	0.13	0.12	0.18	0.24	**0.26**	0.2
7	0.21	**0.25**	0.3	-0.05	0.08	0.04
8	0.51	-0.35	0.06	0.26	**0.31**	0.25
9	0	0	0	-0.01	-3.16	**0.06**

TABLE II

CCC FOR EACH STORY WITH DIFFERENT ADDRESSING VALUES WITH REW MODEL.

	n					
Subject	5	10	15	20	25	30
2	0.09	0.08	-0.06	-0.29	**0.61**	0.35
3	-0.02	-0.01	0.11	0.17	0.35	**0.38**
4	-0.2	-0.19	0.08	0.19	0.44	**0.45**
6	0.09	0.12	0.22	0.24	**0.25**	0.2
7	**0.26**	0.25	-1	-0.06	0.07	0.03
8	-0.31	-0.37	0.24	0.27	**0.31**	0.25
9	0	0	-0.01	-0.02	-1	**0.06**

TABLE III

CCC FOR EACH STORY WITH DIFFERENT ADDRESSING VALUES WITH CREW MODEL.

Concerning the preprocessing methods that were employed, what can be inferred from the validation with a single network is that:

I. Otsu performed much faster than other methods, both in training and in testing, due to the reduced number of steps of its algorithm; II. Gaussian adaptive filter and Otsu's binarization had their performance over time increased considerably as n-tuple size increased; III. ReW's performance degenerated faster than that of CReW with Otsu's binarization; IV. ReW's performance increases faster as n-tuple size increases, V. Canny filter performed considerably better over test time than training time; and VI. Sauvola method performed slower than the other preprocessing schemes in all cases.

The results of the ensembles in the Validation set are in Table IV, all ensembles achieved CCC tending to zero, regardless of the preprocessing used. Bagging ReW was also tested in the test set, with virtually identical results.

Ensemble	Binarization	Training (s)	Validation (s)	CCC
Bagging	Gaussian	1.56	7523.65	0
	Sauvola	1.31	7396.88	9.93*e-05
	Canny	2.23	7481.77	-0.0003
	Otsu	1.15	7453.75	0.0001
Na\"ive	Gaussian	3.13	7571.90	0
	Sauvola	2.99	7678.32	-1.93*e-05
	Canny	1.34	7421.09	9.73*e-06
	Otsu	2.7	7459.93	-6.19*e-06

TABLE IV

CCC FOR EACH BAGGINGREW AND NAÏVE REW ENSEMBLE IN VALIDATION SET OF PERSONALIZED TRACK

Individual neural networks outperforms their ensembles in validation, probably because the number of weak learners was not sufficiently representative. The lack of bootstrap and pruning algorithms for Bagging ReW probably caused the reduced learning of each learner to generate noise propagated by the ensemble, instead of efficiently combining the specialized learning of each weak learner. However, ReW and CReW used multimodal learning, while the ensembles used information provided only by the face image, so we see here the impact that adding more modes in this type of task.

V. CONCLUSION

The motivation of this work is to explore the use of weightless neural networks and their ensembles in tasks of affective prediction in video contexts, something unprecedented and extremely important, due to the high speed of training of these models in relation to the state-of-the-art and similar ones, without compromising their competitiveness. The results here are preliminary, turning more to an exploration of the different types of image preprocessing in the multi-modal context, with a single neural network. Another object of study here is the uni-modal prediction for ensembles, where a committee of neural networks is able to circumvent any flawed predictions.

Among the weightless models, ReW obtained the best result, with a CCC of 0.25. While the state-of-the-art for this task is 0.17 in the Test Set, we are unaware of the state-of-the-art for the Validation Set, which we use here, but this is a relatively new task and with many issues still unexplored, so there is a long way to really obtain an accurate prediction of affective states in multi-modal environments, and we believe that WiSARD, given its agility, can be a useful resource in this endeavor, as long as there is the correct treatment for the data.

Due to the time issue, many validations that have not yet been completed have not been reported here. Ongoing works include testing the Boost ReW, varying the hyper parameters of the ensembles, mainly using high values for the number of learners, using audio as an input for the ensembles, using dialogue transcript and body language as alternatives modes, using multi-modal learners in the ensembles, using ensembles with different uni and multi-modal learners and a appropriate comparison with state-of-the-art solutions.

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